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A ZeroTier network is essentially a secure Local Area Network (LAN) that you can use anywhere in the world. Let's make one and connect two devices over ZeroTier.

We'll use `ping` to test the connection. Any two devices that can run ZeroTier will do: laptop, phone, virtual machine, etc...

Both devices can be at the same location, on the same physical network. If you move one to a cafe or to your office, it should still just work.

The rough outline is:

- Create a ZeroTier network
- Join the network from two devices
- `ping` one device from the other over the ZeroTier network

This should take about 5 minutes.

▸ Details

Create your first ZeroTier network

Create an account

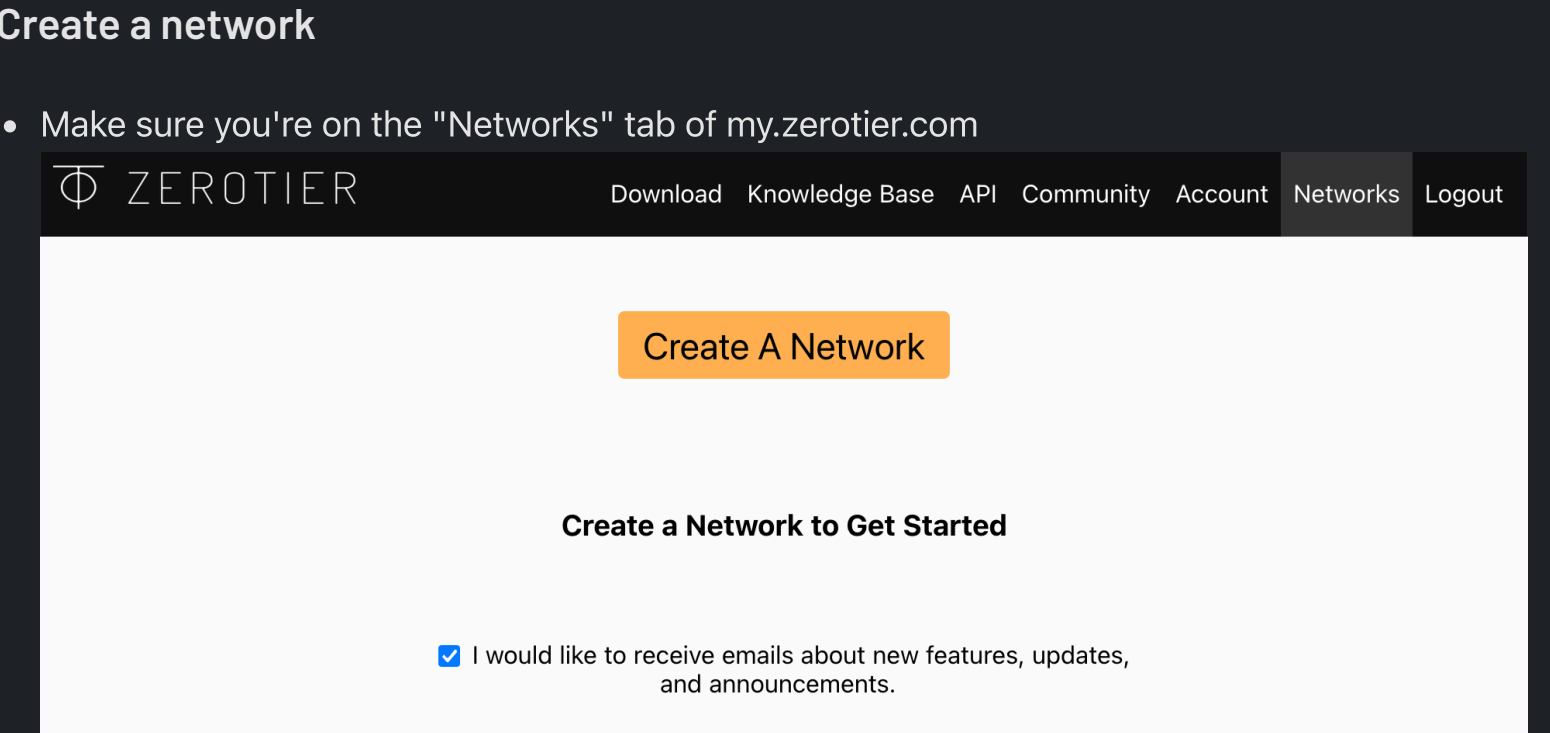
NOTE

It's free, no credit card is required.

- Go to my.zerotier.com and create an account.

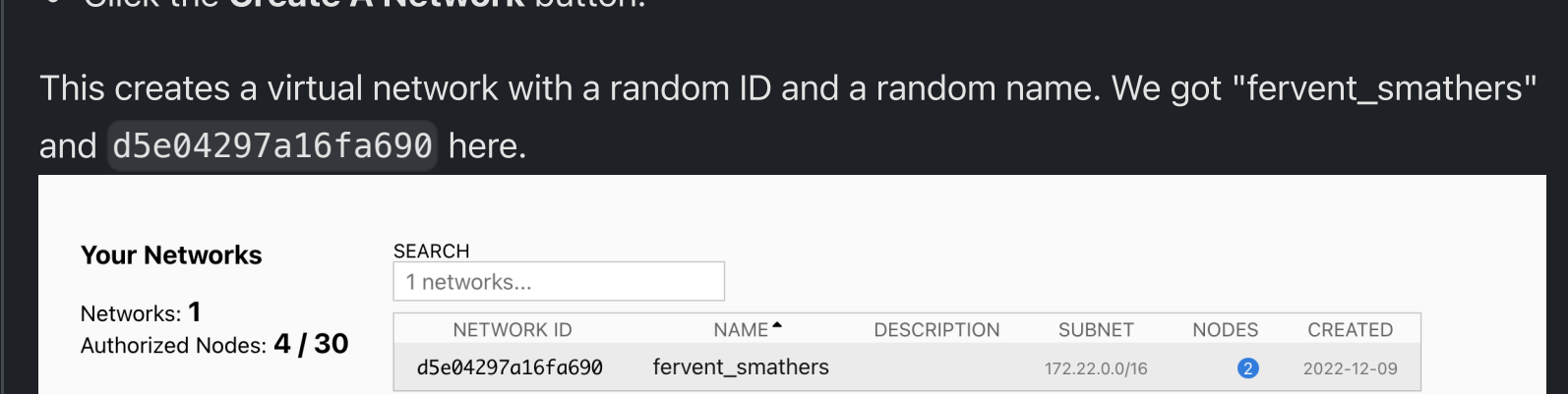
Create a network

- Make sure you're on the "Networks" tab of my.zerotier.com


ZEROTIER
Download Knowledge Base API Community Account Networks Logout
Create A Network
Create a Network to Get Started
☑ I would like to receive emails about new features, updates, and announcements.

- Click the **Create A Network** button.

This creates a virtual network with a random ID and a random name. We got "fervent_smathers" and `d5e04297a16fa690` here.

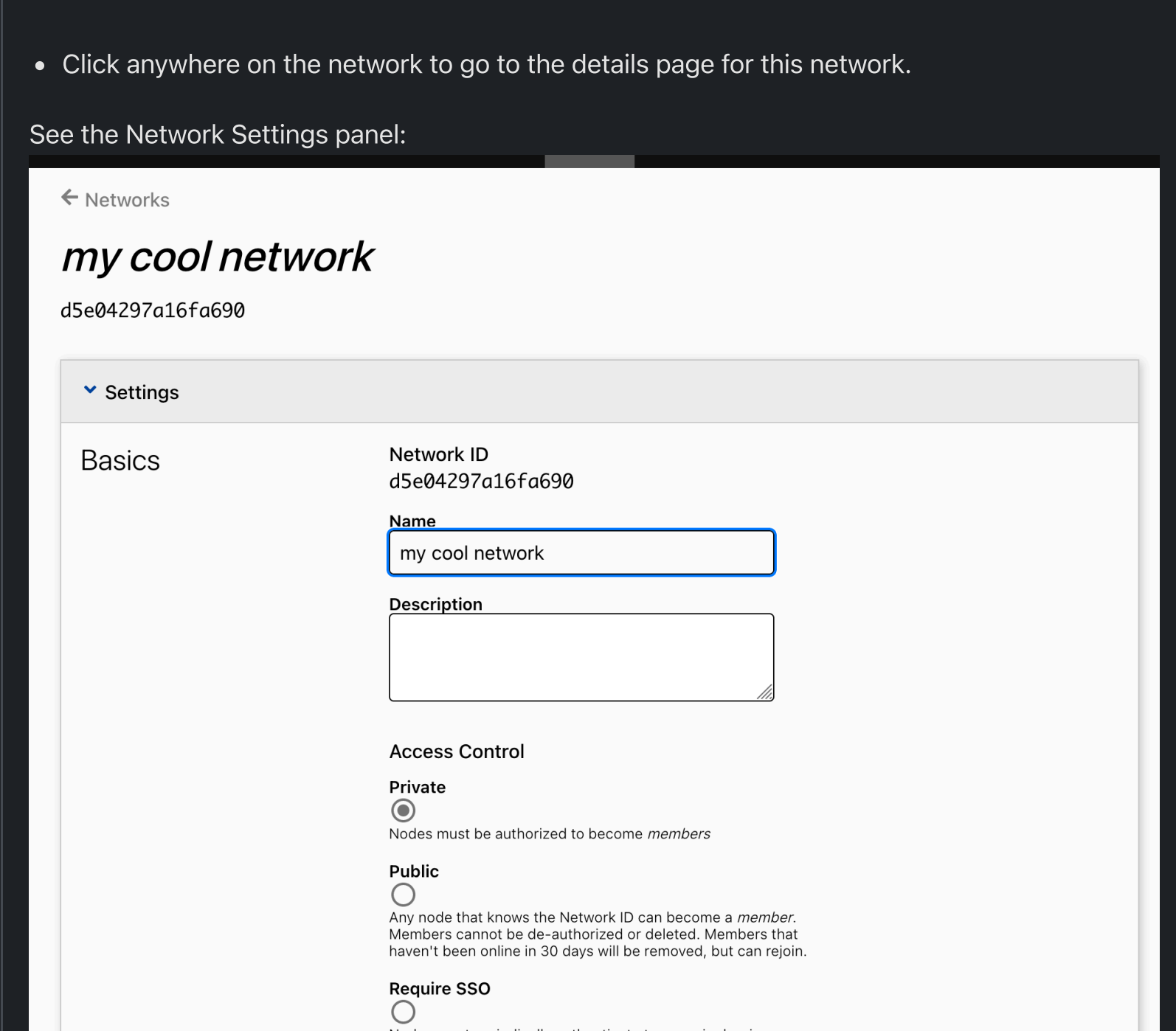


Your Networks
Networks: 1
Authorized Nodes: 4 / 30
SEARCH
1 networks...

NETWORK ID	NAME	DESCRIPTION	SUBNET	NODES	CREATED
d5e04297a16fa690	fervent_smathers		172.22.0.0/16	2	2022-12-09

- Click anywhere on the network to go to the details page for this network.

See the Network Settings panel:



my cool network
d5e04297a16fa690

Settings
Basics
Network ID
d5e04297a16fa690
Name
my cool network
Description

Access Control
Private ●
Nodes must be authorized to become *members*
Public ○
Any node that knows the Network ID can become a *member*. Members cannot be de-authorized or deleted. Members that haven't been online in 30 days will be removed, but can rejoin.
Require SSO ○
Nodes must periodically authenticate to your single-sign-on identity provider to join this network. Clients must be running version 1.10.0 or above

We don't need to change any settings, but we can change the name of the network to personalize it.

- Change "fervent_smathers" to "my cool network" or whatever you like.
- Collapse the Settings panel. Click on the word "Settings" at the top of the panel.

You don't need to change any other settings.

- See the Network Members panel:

It should say "No devices have joined this network".

- Leave this browser tab open. We'll look at it again later.

Setup the ZeroTier app

Download and install ZeroTier

For mobile devices, use the app store.

- Go to zerotier.com/download in a different tab of your browser.
- Run the installer

The ZeroTier client should now running on your device.

Join your first ZeroTier network

We need to tell the client to "join" the virtual network we just created.

- Copy the Network ID of the network from my.zerotier.com This is the long number that looks like like: `d5e04297a16fa690`
- Paste the Network ID into the "join" command on your device

On macOS and Windows, there is a menubar/tray app. Select "join" from the menu.

macOS Windows Command Line Mobile

NOTE

Every running instance of ZeroTier has a unique address. It's the 10 digit "Address" in the app, or `zerotier-cli info` command.

ZeroTier addresses are a very secure method of unique identification.

Authorize your device on your network

At this point, your client should say "Access Denied." A device can't talk on your network unless you allow it, even if someone discovers the network's ID.

macOS Windows Linux Mobile

Authorize your device

- Go to the Members panel that we left open on my.zerotier.com
- Your node that just "joined" should appear here.
- The "Address" should match the address in your client.
- Click the "Auth?" check box for it.
- Give it a name. Type something like "laptop" or "bob" into the `(short name)` input.

Confirm authorization

Back on your computer, your client should now say "OK" instead of "ACCESS DENIED" and it should show your custom "my cool network" name.

macOS Windows Linux Mobile

Now you have 1 member on your network. A network with 1 member can't do much.

Repeat with another device

We need to have 2 devices connected to the same ZeroTier network.

- Repeat the join and authorize steps with your second device.

Test connectivity

Now you have two authorized nodes on your network. They should be able to talk over ZeroTier.

Your Network Members section should look something like this:

The "Managed IPs" will be different on your network.

We're going to test with `ping`. It's the only program that we can think of that exists by default on every operating system.

This is a command line program, but don't worry: You can do it.

Gotcha: Windows blocks ping

~~Windows by default doesn't respond to pings. If you try to ping a Windows computer from a different computer, it won't work. You can enable ping.~~

ZeroTier automatically enables ping on your ZeroTier network adapter now. You can probably skip this step!

▸ How to enable ping on Windows

Open the command line

- Open the command line on your computer

macOS Windows Linux Mobile

- Use Spotlight (cmd-space) to search for Terminal

Apple's Instructions

Find the ZeroTier IP Addresses of your devices

Try the ping command

For macOS and Linux users, you are going to do the following.

Back in the Command Line / Terminal that you just opened:

- type `ping -c 5 $ZEROTIER_IP_ADDRESS` `<enter>` into your command line.

A successful `ping`:

```
% ping -c 5 172.22.217.93
PING 172.22.217.93 (172.22.217.93): 56 data bytes
64 bytes from 172.22.217.93: icmp_seq=0 ttl=64 time=22.362 ms
64 bytes from 172.22.217.93: icmp_seq=1 ttl=64 time=10.157 ms
64 bytes from 172.22.217.93: icmp_seq=2 ttl=64 time=9.414 ms
64 bytes from 172.22.217.93: icmp_seq=3 ttl=64 time=9.019 ms
64 bytes from 172.22.217.93: icmp_seq=4 ttl=64 time=9.180 ms

--- 172.22.217.93 ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 9.019/12.026/22.362/5.182 ms
```

Try it with both ZeroTier Managed addresses on your network.

One of them is the same device you're on, so you're pinging yourself. Pinging the other device might be a little more interesting.

INFO

If something goes wrong you might see something like:

```
% ping -c 5 172.22.217.92
PING 172.22.217.92 (172.22.217.92): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3

--- 172.22.217.92 ping statistics ---
5 packets transmitted, 0 packets received, 100.0% packet loss
```

or

```
ping -c 5 192.168.123.234
PING 192.168.123.234 (192.168.123.234): 56 data bytes
92 bytes from 192.168.82.1: Destination Port Unreachable
Vr HL TOS  Len   ID Flg  off TTL Pro  cks      Src      Dst
 4  5  00 5400 56e7   0 0000  3f  01 d4ad 192.168.82.217  192.168.123
```

For Windows users, the `-c` parameter is explained as something else. If you want to send 5 packages, you will do the following.

- type `ping -n 5 $ZEROTIER_IP_ADDRESS` `<enter>` into your command line.

A successful `ping`:

```
C:\> ping -n 5 172.22.217.93
Pinging 172.22.217.93 with 32 bytes of data
Reply from 172.22.217.93: bytes=32 time=22ms TTL=64
Reply from 172.22.217.93: bytes=32 time=10ms TTL=64
Reply from 172.22.217.93: bytes=32 time=9ms TTL=64
Reply from 172.22.217.93: bytes=32 time=7ms TTL=64
Reply from 172.22.217.93: bytes=32 time=9ms TTL=64

Ping statistics for 172.22.217.93:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 7ms, Average = 6ms
```

There may just be a typo in the IP address. Double check that your device is authorized at my.zerotier.com

Contact us on the discussion forum and see the troubleshooting section if you get stuck.

Conclusion

`ping` doesn't accomplish anything, but it does tell us ZeroTier is working. It's useful to know about for troubleshooting networks, not just ZeroTier networks.

Visit the discussion forum to talk about your use-cases or if you get stuck.

Leaving a Network

When you no longer need to be connected to a network, you can leave it:

macOS and Windows

- Click the ZeroTier icon in your menu bar/system tray
- Click the network name to uncheck it (or use **Network Details** to uncheck the Connected checkbox)
- To permanently remove a network, use **Network Details** -> **Delete Network**

Command Line

```
sudo zerotier-cli leave NETWORK_ID
```

Now, use ZeroTier to do something you want to do

Some popular uses

- Windows Remote Desktop
- ssh (try mosh)
- Private Gaming LAN
- Access the web interfaces of your home lab
- Build your own VPN
- Route to a remote subnet
- Route to a Docker network
- Add dns to your network

Join multiple networks

A node can join many networks at once. Make sure they don't use the same subnet!

You can have a `home` network, a `friends` network, and a `work` network, for example.

They don't all need to be networks that you've created. You can join other people's networks.

Check out the Whitepaper

For more info on the cryptography and protocol, see the: Design Whitepaper

Tags: tutorial beginner installation client central admin

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